

## **Initial Post**

by Jens Kolby - Friday, 1 August 2025, 2:39 PM

Number of replies: 2

Over the past decades, computing has significantly transformed industries, individuals, and societies, reshaping how businesses operate and interact globally. It took several high-profile incidents before industries acknowledged its importance and adapted their practices. One of those cyberattacks was the Morris Worm (1988), which infected thousands of UNIX machines and disrupted significant portions of the early Internet. According to Brassard (2018, p. 8), this incident was a catalyst for the creation of the Computer Emergency Response Team (CERT), established to coordinate future responses to cybersecurity threats. Lessons learned from these early cybersecurity incidents laid the foundation for today's cybersecurity practices, which have become essential in modern industries, especially those handling sensitive data, such as the financial sector.

This impact has driven the financial industry to adopt regulatory frameworks such as the EU's Digital Operational Resilience Act (DORA), implemented in 2024 (ISACA, 2024), and SWIFT's Customer Security Programme (CSP) (ISACA, 2022), both aimed at strengthening cyber defenses and enforcing mandatory controls. CSP was introduced following the 2016 Bangladesh Bank cyberheist (Sultana, 2024), in which attackers compromised SWIFT terminals via spear-phishing, resulting in significant monetary losses. These developments have not only strengthened internal banking operations but

also improved customer protection worldwide. This evolution is part of a constantly changing cybersecurity landscape, with recent examples demonstrating that it continues to play a critical role in shaping both business and society.

The rapid advancement of machine learning in the financial sector has transformed both opportunities and risks. New vulnerabilities are emerging that traditional cybersecurity methods cannot always effectively defend against. According to Zou et al. (2022), ML systems require targeted security analysis, as classical security techniques are often insufficient to identify and mitigate threats aimed specifically at ML models. This development highlights that modern cyberattacks increasingly exploit AI and ML, creating a constant need for tailored defense strategies within the financial industry and the wider digital economy.

Grammarly and ChatGPT were used solely for planning, proofreading, and language enhancement. All academic content, reasoning, and conclusions are the author's own.

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